

Cover Letter — Submission to Progress of Theoretical and Experimental Physics (PTEP)

To the Editors of *Progress of Theoretical and Experimental Physics*,

I am pleased to submit the manuscript “**Hyperbolic Flavor Geometry: Mixing, CP Violation, and Fermion Masses from Arithmetic Hyperbolic 3-Manifolds**” for consideration as a research article in PTEP.

Summary. The paper establishes three geometric theorems about two compact arithmetic hyperbolic 3-manifolds— $M_{\text{PMNS}} = m003(-2, 3)$ (the Meyerhoff manifold, minimum-volume closed orientable hyperbolic 3-manifold) and $M_{\text{CKM}} = m006(-5, 2)$ —and then presents computational observations showing that their holonomy representations encode Standard Model flavor parameters with no continuously tuned free parameters:

- The surgery formula $|H_1(m003(p, q))| = 5|2p + q|$, proved from the fundamental group presentation, gives $H_1(M_{\text{PMNS}}) = \mathbb{Z}/5$ as a theorem.
- The cusp shape $\tau = e^{i\pi/3}$ is exact, following from the imaginary quadratic trace field $\mathbb{Q}(\sqrt{-3})$.
- The unique degree-5 algebraic covering space of M_{PMNS} introduces new torsion prime $11 = L_5$ (fifth Lucas number), equal to the Farey intersection of the two HFG filling slopes.

Physical results: the PMNS mixing matrix (Frobenius fitness 0.005087, $p < 10^{-4}$), the leptonic CP phase $\delta = 195.91^\circ$ (0.55% from PDG 2024, zero free parameters), and charged fermion mass ratios as arithmetic norms in $\mathbb{Z}[\omega]$ and $\mathbb{Z}[\sqrt{17}]$ ($p < 0.002$ by permutation test). The sign of the trace field discriminant (-3 vs $+17$) explains the qualitative lepton/quark CP asymmetry geometrically.

Relevance to PTEP. PTEP has a strong tradition of publishing geometric and algebraic approaches to flavor physics, including papers on modular flavor symmetry, extra dimensions, and non-Abelian discrete groups. The present work complements this tradition by grounding flavor structure in the arithmetic geometry of hyperbolic 3-manifolds—compact objects with fully rigorous mathematical invariants (volume, trace field, covering tower)—and presenting the Standard Model correspondences as falsifiable computational observations. The leptonic CP phase prediction ($\delta = 195.91^\circ$) is directly testable by Hyper-Kamiokande and DUNE. PTEP’s SCOAP3 open-access framework makes it the ideal venue for this work, which bridges the mathematics and physics communities.

Companion papers. A companion paper establishing the pure mathematical results (“A Quadratic Cover Prime Formula for a Farey Tower of Arithmetic Hyperbolic 3-Manifolds,” SSRN 6808878) is under review at *Experimental Mathematics*. Papers on the individual components—CKM mixing (J. Geom. Phys., JGP13076) and PMNS mixing (J. Geom. Phys., JGP13101)—are under concurrent review. The present paper is the unified framework; all results are independently reproducible.

Reproducibility. All computations use SnapPy at 150-bit precision. Canonical reproduction scripts are publicly available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>; the main results reproduce in under 60 seconds on any standard laptop.

Suggested referees.

- Prof. Nathan Dunfield (University of Illinois), nmd@illinois.edu — computational hyperbolic geometry and SnapPy.
- Prof. Alan Reid (Rice University), areid@math.rice.edu — arithmetic hyperbolic 3-manifolds and trace fields.
- Prof. Ferruccio Feruglio (University of Padova), feruglio@pd.infn.it — discrete flavor symmetries and lepton mixing.
- Prof. Mu-Chun Chen (UC Irvine), muchunc@uci.edu — lepton mixing, CP violation, and geometric approaches to flavor.

I confirm that this manuscript has not been submitted to any other journal, that there are no competing interests, and that no external funding was received. The research received no institutional support.

I look forward to the opportunity to contribute to PTEP.

Sincerely,

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